

Practical No 7

```
#include <iostream>
#include <queue>

using namespace std;

class CircularQueue {
    int q[5], front, rear;

public:
    CircularQueue() {
        front = rear = -1;
    }

    void enqueue(int call) {
        if ((rear + 1) % 5 == front) {
            cout << "Call queue is full!\n";
            return;
        }

        if (front == -1)
            front = 0;

        rear = (rear + 1) % 5;
        q[rear] = call;

        cout << "Call " << call << " added.\n";
    }

    void dequeue() {
        if (front == -1) {
            cout << "No normal calls waiting.\n";
            return;
        }

        cout << "Handling Call " << q[front] << endl;

        if (front == rear)
            front = rear = -1;
        else
            front = (front + 1) % 5;
    }
};

int main() {
    CircularQueue calls;
    priority_queue<int> priorityCalls;

    calls.enqueue(101);
```

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calls.enqueue(102);
calls.enqueue(103);

priorityCalls.push(201);
priorityCalls.push(203);
priorityCalls.push(202);

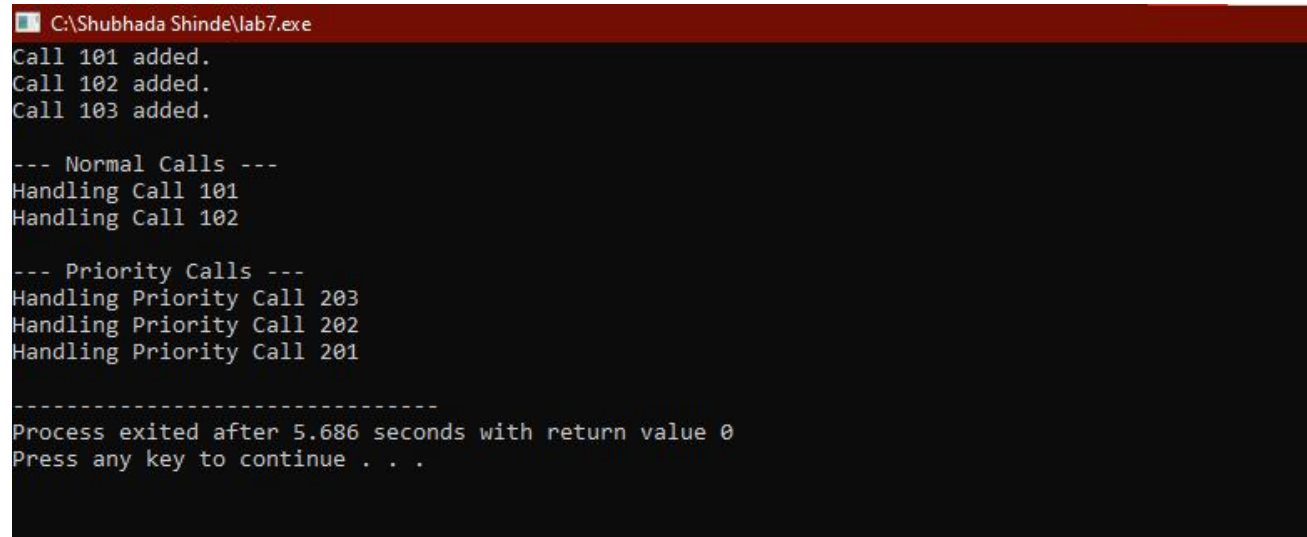
cout << "\n--- Normal Calls ---\n";
calls.dequeue();
calls.dequeue();

cout << "\n--- Priority Calls ---\n";
while (!priorityCalls.empty()) {
    cout << "Handling Priority Call "
        << priorityCalls.top() << endl;
    priorityCalls.pop();
}

return 0;
}

```

Output:



```

C:\Shubhada Shinde\lab7.exe
Call 101 added.
Call 102 added.
Call 103 added.

--- Normal Calls ---
Handling Call 101
Handling Call 102

--- Priority Calls ---
Handling Priority Call 203
Handling Priority Call 202
Handling Priority Call 201

-----
Process exited after 5.686 seconds with return value 0
Press any key to continue . . .

```

